Year 1 AS Pure Maths

Chapter 3

Equations and inequalities

Practice Booklet

Contents

3.1 Linear simultaneous equations ..... book p. 16
3.2 Quadratic simultaneous equations ..... book p. 17
3.3 Simultaneous equations on graphs ..... book p. 18
3.4 Linear inequalities ..... book p. 19
3.5 Quadratic inequalities ..... book p. 20
3.6 Inequalities on graphs ..... book p. 21
3.7 Regions ..... book p. 22
Problem solving: Set A ..... book p. 24
Problem solving: Set B ..... book p. 25
Selected exam question bank questions .....
Answers .....

3.1 Linear simultaneous equations

Practice Book, printed page 16.

1. Solve these simultaneous equations by elimination:

a. $6x + y = 9$
$4x - y = 11$

b. $2x + 3y = 8$
$3x + 2y = 7$

c. $4x - 3y = 2$
$5x - 7y = 9$

Hint

Either the x- or y-coefficients in the pair of equations need to have the same value.
You can then add or subtract the equations to eliminate one variable and solve for the other.
You need to find the value of both variables.

2. Solve these simultaneous equations by substitution:

a. $5x - 2y = 3$
$x + 4y = 5$

b. $2x + 5y = 37$
$y = 11 - 2x$

c. $4x + 3y = 5$
$2x - 6y = -5$

Hint

Rearrange one equation to make x or y the subject and then substitute this expression into the other equation and solve.

3. Solve these simultaneous equations:

a. $3(x - y) + 6 = 0$
$y + x = 8$

b. $5(x - 1) = -2y$
$3x - 29 = 4y$

c. $\frac{4x - y}{2} = 11$
$\frac{5 - 3x}{5} = y$

Hint

Rearrange the equations into the form $ax + by = c$, where a, b and c are constants and then solve using either elimination or substitution.

4. E Solve the simultaneous equations
$x + y = 2$
$2y = 18x - 6$ (2 marks)

5. E Solve the simultaneous equations
$3x - y = -5$
$0.5y + 2x = 4$
giving your answers in exact form. (2 marks)

6. E/P $6ky + 9x = 12$
$ky - x = 4.5$
are simultaneous equations where k is a constant.

a. Show that $x = -1$. (2 marks)

b. Given that $y = 7$, find the value of k. (1 mark)

7. E/P Two students are attempting to solve the simultaneous equations
$4x + 6y = 10$
$2x = 5 - 3y$
Ben says that these equations have no solutions, and Nisha says that they have infinitely many solutions. Who is correct? Explain your answer. (2 marks)

3.2 Quadratic simultaneous equations

Practice Book, printed page 17.

1. Solve the simultaneous equations:

a. $xy = 64$
$4x - y = 60$

b. $x^2 + y^2 = 10$
$x + y = 4$

c. $x - y + 1 = 0$
$3x^2 - 4y = 0$

Hint

A quadratic equation can contain terms involving xy, x^2 and y^2.
Rearrange the linear equation to make x or y the subject, then substitute this expression into the quadratic equation and solve.

2. Solve the simultaneous equations:

a. $y - x^2 + 3x + 2 = 0$
$y - 2x + 6 = 0$

b. $x^2 + 2x - y = 14$
$y + 2 = x$

c. $x^2 + y^2 = 5$
$y = 5 - 3x$

Hint

Each set of equations will have two pairs of solutions.
You need to identify the solutions and pair them up correctly.

3. Solve these simultaneous equations, giving your answers to 2 decimal places:

a. $3x - 7 = y$
$x^2 - 3x - 2 = 2y$

b. $2x^2 - xy + y^2 = 8$
$x + y = 1$

c. $y^2 - 5x^2 = 20$
$4x - 7 = y$

4. E Solve the simultaneous equations
$x + y = 4$
$4y^2 - x^2 = 12$ (4 marks)

5. E Solve the simultaneous equations
$y + 2x + 1 = 0$
$y^2 + 3x^2 + 2x = 0$ (4 marks)

6. E/P $mx - y - 2 = 0$
$x^2 - 2x + y^2 - 4y = 4$
where m is a real constant.
Given that these simultaneous equations have exactly one pair of solutions, find the two possible values of m, giving your answers in exact form. (7 marks)

7. E/P The values of x and y satisfy the simultaneous equations
$y - 2x = 8$
$x^2 + 2ky + 4k = 0$
where k is a non-zero constant.

a. Show that $x^2 + 4kx + 20k = 0$. (2 marks)

b. Given that $x^2 + 4kx + 20k = 0$ has equal roots, find the value of k. (3 marks)

c. For the value of k found in part **b**, solve the simultaneous equations. (3 marks)

3.3 Simultaneous equations on graphs

Practice Book, printed page 18–19.

1. Sketch the graphs of each pair of equations on the same set of axes. In each case, find the coordinates of the point of intersection.

a. $2x - y = 8$
$4x + 3y = 6$

b. $2x - 5y = 5$
$3x - 2y = 13$

Hint

You can sketch straight lines in the form $ax + by = c$ quickly by setting $x = 0$ and solving to find where the line crosses the y-axis, and similarly setting $y = 0$ and solving to find the points where the line crosses the x-axis.

2.

a. On the same set of axes, sketch the curve with equation $y = x^2 - 8x + 15$ and the line with equation $y = 3x - 3$.

b. Find the coordinates of the points where the line and the curve intersect.

c. Verify your solutions to part **b** are correct by substitution.

Hint

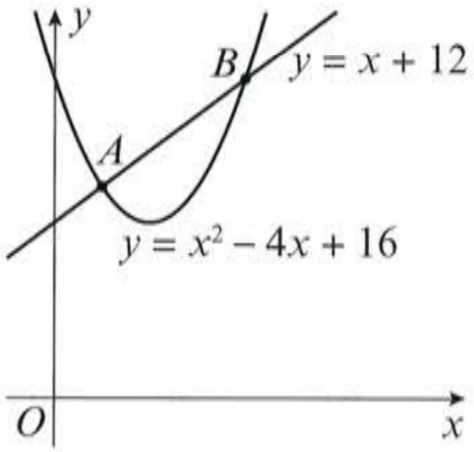
First label the points where the curve and the line cross the coordinate axes. Then sketch both graphs.

3.

a. On the same set of axes, sketch the curve with equation $y = 3x^2 + 8x - 3$ and the line with equation $y = 7x - 2$.

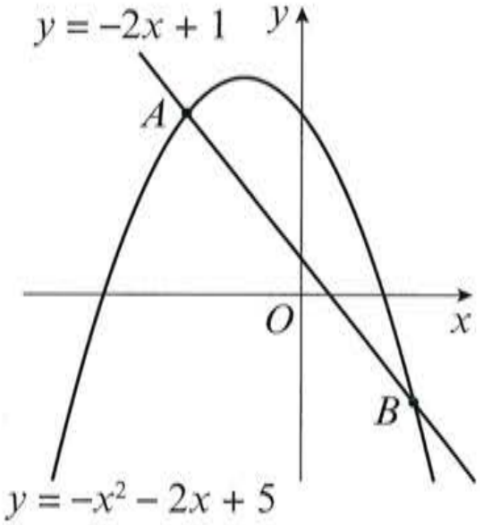
b. Find the coordinates of the points of intersection, giving your answers to 2 decimal places.

4. E The diagram shows the curve with equation $y = x^2 - 4x + 16$ and the line with equation $y = x + 12$. The curve intersects the line at points A and B.



Using an appropriate algebraic method, find the coordinates of A and B. (4 marks)

5. E The diagram shows part of curve with equation $y = -x^2 - 2x + 5$ and part of the line with equation $y = -2x + 1$. The curve intersects the line at points A and B.



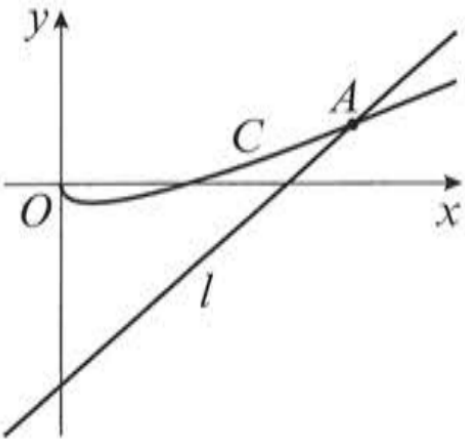
Using an appropriate algebraic method, find the coordinates of A and B. (4 marks)

6. E/P $p(x) = 1 - \frac{1}{2}x$, $q(x) = x^2 - 4x - 10$

a. Sketch the graphs of $y = p(x)$ and $y = q(x)$ on the same set of axes. Label all points where the graphs cut the coordinate axes. (4 marks)

b. Use an algebraic method to find the coordinates of any points of intersection of the graphs $y = p(x)$ and $y = q(x)$. (4 marks)

7. E/P The diagram shows a sketch of the curve C with equation $y = 2x - 3\sqrt{x}$ $(x \geqslant 0)$, and the line l with equation $y = 3x - 12$.



The line cuts the curve at point A as shown in the diagram.
Using algebra, find the x-coordinate of point A. (5 marks)

8. E/P $f(x) = 6 - 5x$ and $g(x) = 4 - 0.5x$

a. Use an algebraic method to find the exact coordinates of the point of intersection of the graphs of $y = f(x)$ and $y = g(x)$. (3 marks)

b. Sketch these graphs on the same set of axes, showing all intersections with the coordinate axes. (2 marks)

3.4 Linear inequalities

Practice Book, printed page 19–20.

1. Find the set of values of x for which:

a. $x + 4 \geqslant 12$

b. $7 - 5x < 62$

c. $3(3 - 2x) \leqslant 2(3 + 2x)$

Hint

You can solve inequalities like equations. However, if you divide both sides by a negative number, you need to reverse the inequality sign.

2. Find the set of values of x for which:

a. $7 \leqslant 5x - 3 \leqslant 17$

b. $\frac{x}{3} - \frac{2x}{9} < 3$

c. $3x(3 + x) + x^2 \geqslant 1 + x(6 + 4x)$

Hint

For part **c**, expand the brackets and simplify before solving.

3. Use set notation to describe the set of values of x for which:

a. $2(1 + x) < 4 - x$ and $x - 3 < 4x + 6$

b. $\frac{x}{4} + 3 \leqslant 6$ and $2(3x - 5) \geqslant 20$

c. $0.5(4x + 3) < 2.5$ or $\frac{x - 3}{5} > 7$

Hint

In set notation $x < 3$ and $x > -1$ is written as $\{x : -1 < x < 3\}$, or alternatively $\{x : x > -1\} \cap \{x : x < 3\}$.

4. E Find the set of values of x for which:

a. $2x - 6 > 3 - x$ (2 marks)

b. $4x - 5 < 3 + 2x$ (2 marks)

5. E Use set notation to write the set of values of x for which:

a. $20 - 2x > 15 - 7x$ (2 marks)

b. $x - 5 \leqslant 3 - 9x$ (2 marks)

6. E Use set notation to write the set of values of x for which:

a. $3(3x + 4) \geqslant 2 - x$ (2 marks)

b. $7(x - 3) < 3(x - 3)$ (2 marks)

7. E/P The width of a rectangular field is x metres, $x > 0$. The length of the pitch is 30 m more than its width. Given that the perimeter of the pitch must be less than 400 m,

a. form a linear inequality in x (2 marks)

b. solve your inequality. (2 marks)

8. E/P Use set notation to describe the set of values of x for which:

a. $3(2 + x) \geqslant 2 - x$ and $x + 7 > 5x - 1$ (4 marks)

b. $0.25(8x + 4) < 5.5$ or $\frac{x - 2}{9} > 11$ (4 marks)

3.5 Quadratic inequalities

Practice Book, printed page 20–21.

1. Find the set of values of x for which,

a. $x^2 < 25$

b. $x(2x + 1) < 3$

c. $0 > 1 - 3x^2 - 2x$

d. $2x^2 \geqslant 5x$

Hint

First, replace the inequality sign by an equals sign and solve the quadratic equation to find the critical values. Then use a sketch of the graph to find the required set of values.

2. Use set notation to describe the set of values of x for which:

a. $x^2 - 5x - 14 < 0$ and $5x + 2 > 12$

b. $2x^2 - 3x - 5 > 0$ and $x^2 + 3x - 4 < 0$

c. $3x^2 + 12x - 15 < 0$ and $x^2 + 5x + 6 > 0$

Hint

Draw number lines to illustrate the pairs of inequalities.

3. Given that $x \neq 0$, find the set of values of x for which:

a. $36 > \dfrac{1}{x^2}$

b. $3 < \dfrac{5}{x}$

c. $16 + \dfrac{6}{x} \geqslant \dfrac{2}{x}$

Hint

Start by multiplying both sides of the inequality by x^2.

4. E Show that $x^2 - 6x + 11 > 0$ for all real values of x. (3 marks)

5. E/P Find the set of values of x for which:

a. $2x - 2 > 7 - x$ (2 marks)

b. $x^2 - 6x \leqslant 40$ (4 marks)

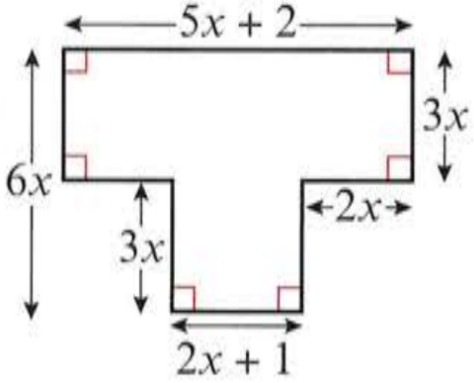
c. **both** $2x - 2 > 7 - x$ **and** $x^2 - 6x \leqslant 40$ (1 mark)

6. E/P The equation $x^2 + (k - 2)x + (4 - 2k) = 0$, where k is a constant, has two distinct real roots.

a. Show that k satisfies $k^2 + 4k - 12 > 0$ (3 marks)

b. Find the set of possible values of k. (4 marks)

7. E/P The diagram shows the plan of a park. The edges are straight lines and the lengths shown in the diagram are given in metres.



a. Given that the perimeter of the park is greater than 103 m, show that $x > 4.5$. (3 marks)

b. Given that the area of the park is less than 702 m^2, form and solve a quadratic inequality in x. (5 marks)

c. Hence state the range of the possible values of x. (1 mark)

8. E/P The equation $2x^2 - 3kx + k = 0$ (where k is a constant) has no real roots. Find the set of possible values of k. (4 marks)

9. E/P The curve C has equation $y = x^2 + 3px + 8p$, where p is a real constant. The straight line L has equation $y = 1 - 2x$. Given that L does not intersect C, find the set of possible values of p, giving your answers in exact form. (8 marks)

10. E/P Given that $x \neq 1$, find the set of values for which $\dfrac{3}{x-1} < 2$ (6 marks)

3.6 Inequalities on graphs

Practice Book, printed page 21–22.

1. For each pair of functions,
i sketch the graphs of $y = f(x)$ and $y = g(x)$ on the same set of axes and find the coordinates of the point of intersection
ii write down the solution to the inequality $f(x) < g(x)$.

a. $f(x) = 8 - 3x$
$g(x) = 7x - 2$

b. $f(x) = 2x - 3$
$g(x) = 5x + 11$

c. $f(x) = 6 - 2x$
$g(x) = 9 - 5x$

Hint

The values of x for which the curve $y = f(x)$ is below the curve $y = g(x)$ satisfy the inequality $f(x) < g(x)$.

2. For each pair of functions,
i sketch the graphs of $y = f(x)$ and $y = g(x)$ on the same set of axes and find the coordinates of the points of intersection
ii write down the solutions to the inequality $f(x) \leqslant g(x)$.

a. $f(x) = x^2 + 6$
$g(x) = -2x + 9$

b. $f(x) = 4 - x^2$
$g(x) = 3x + 4$

c. $f(x) = x^2 + x - 12$
$g(x) = -x - 9$

Hint

Set $f(x) = g(x)$, rearrange into a standard quadratic form and then solve to find the points of intersection.

3. For each pair of functions,
i sketch the graphs of $y = f(x)$ and $y = g(x)$ on the same set of axes and find the coordinates of the points of intersection
ii use set notation to write down the solutions to the inequality $f(x) > g(x)$.

a. $f(x) = 2x^2 + 2x$
$g(x) = 5x + 2$

b. $f(x) = -x^2 + 5x + 6$
$g(x) = 3x + 6$

c. $f(x) = x^2 + 3x + 2$
$g(x) = 2 - x$

Hint

The values of x for which the curve $y = f(x)$ is above the curve $y = g(x)$ satisfy the inequality $f(x) > g(x)$.

4. E

a. Sketch the graphs of $f(x) = 3x - 7$ and $g(x) = 3 - x$ on the same set of axes. (2 marks)

b. Find the coordinates of any points of intersection. (2 marks)

c. Find the set of values of x for which $3x - 7 > 3 - x$. (1 mark)

5. E/P

a. Sketch the graphs of $f(x) = 8x - 12 - x^2$ and $g(x) = 2x - 7$ on the same set of axes. Label all points where the curves cut the coordinate axes. (4 marks)

b. Find the coordinates of any points of intersection. (4 marks)

c. Find the set of values of x for which $8x - 12 - x^2 > 2x - 7$ (1 mark)

6. E/P $p(x) = \frac{7}{2}x - \frac{31}{2}$, $q(x) = x^2 - 3x - 10$

a. Solve the equation $q(x) = 0$. (2 marks)

b. Sketch the graphs of $y = p(x)$ and $y = q(x)$ on the same set of axes. Label all points where the curves cut the coordinate axes. (4 marks)

c. Use an algebraic method to find the coordinates of any points of intersection of the graphs $y = p(x)$ and $y = q(x)$. (4 marks)

d. Write down, using set notation, the set of values of x for which $p(x) < q(x)$. (2 marks)

3.7 Regions

Practice Book, printed page 22–23.

1.

a. On a coordinate grid, shade the region satisfying the inequalities
$y \geqslant -3, \quad x \geqslant 0.5 \quad \text{and} \quad y \leqslant -2x + 6$

b. Find the points of intersection of the lines
$y = -3, \quad x = 0.5 \quad \text{and} \quad y = -2x + 6$

c. Find the area of the shaded region.

Hint

If $y \geqslant f(x)$ or $y \leqslant f(x)$, the graph of the function $y = f(x)$ is included in the region and is represented by a solid line.
You can substitute a coordinate point into an inequality to test whether the point lies within the required region.

2. P

a. On a coordinate grid, shade the region that satisfies the inequalities
$y < -(x + 3)(x - 4), \quad y \leqslant x + 8 \quad \text{and} \quad y > -x + 4$

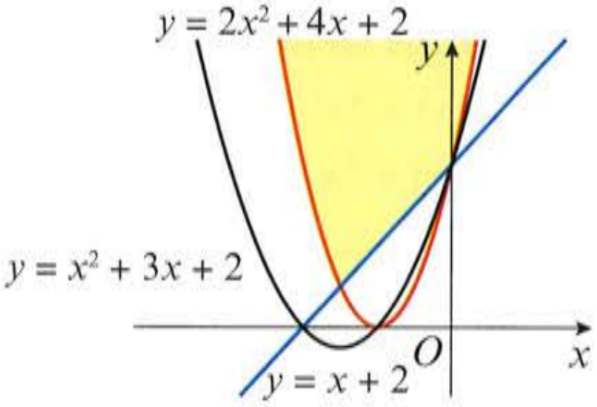
b. Find the coordinates of the vertices of the shaded region.

c. Briefly explain why none of these vertices lies within the shaded region.

Hint

If $y > f(x)$ or $y < f(x)$, the graph of the function $y = f(x)$ is not included in the region and is represented by a dotted line.

3. E/P The sketch shows graphs of the equations
$y = x^2 + 3x + 2, \quad y = 2x^2 + 4x + 2 \quad \text{and} \quad y = x + 2$



a. Find the coordinates of the points of intersection of the graphs. (5 marks)

b. Write down a set of inequalities that define the shaded region. (3 marks)

c. State which, if any, of the points of intersection lie within the region represented by the inequalities. (1 mark)

Hint

Vertices lie within the region only if both intersecting lines are included. Therefore, a vertex must be between two solid lines to be considered within the region.

4. E/P

a. On a coordinate grid, shade the region comprising all points whose coordinates satisfy the inequalities $y \leqslant 3x + 6, \quad 2y + x \leqslant 5 \quad \text{and} \quad y \geqslant \frac{3}{2}$. (3 marks)

b. Work out the area of the shaded region. (4 marks)

5. E/P

a. On a coordinate grid, shade the region comprising all points whose coordinates satisfy the inequalities $y \leqslant 2x + 6$, $4y + 17x \leqslant 4$ and $y \geqslant 1$. (3 marks)

b. Work out the area of the shaded region. (4 marks)

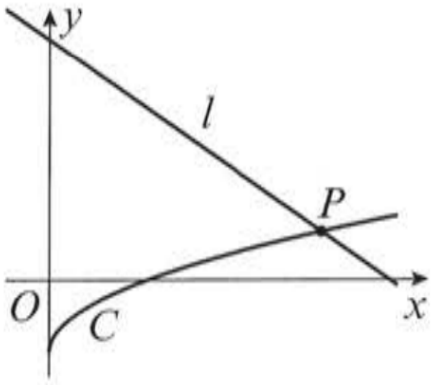
6. E/P

a. On a coordinate grid, sketch the graphs of the equations
$y = x^2 + 3x - 4$, $y = x^2 - x - 2$ and $x = 1$
Label the coordinates of all points where the graphs intersect the coordinate axes. (6 marks)

b. Use an algebraic method to find the coordinates of any point of intersection of the graphs. (6 marks)

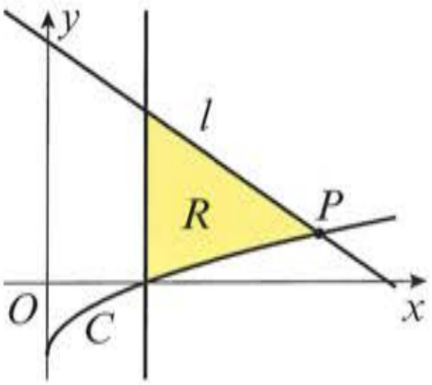
c. On your sketch, shade the region that satisfies the inequalities
$y \leqslant x^2 + 3x - 4$, $y \leqslant x^2 - x - 2$ and $x \leqslant 1$ (1 mark)

7. E/P The diagram shows a sketch of the curve C with equation $y = 2\sqrt{x} - 3$, $x \geqslant 0$, and the line l with equation $y = 10 - 1.28x$



The line cuts the curve at point P.

a. Using algebra, find the x-coordinate of point P. (5 marks)



The region R is enclosed by curve C, the line l, and the vertical line through the point where C meets the x-axis.

b. Identify the inequalities that define R.
The boundary lines are included in the region. (3 marks)

Problem solving: Set A

Bronze

Given the simultaneous equations
$y = 2x + 3$
$x^2 - y + 2k = 0$
where k is a non-zero constant,

a. show that $x^2 - 2x + (2k - 3) = 0$. (2 marks)

Given that $x^2 - 2x + (2k - 3)$ has equal roots,

b. use the discriminant to form an equation, and solve this to find the value of k. (3 marks)

c. For this value of k, solve the simultaneous equations. (3 marks)

Silver

Given the simultaneous equations
$y + \frac{1}{4} = 4x$
$x^2 - 4ky + 3k = 0$
where k is a non-zero constant,

a. show that $x^2 - 16kx + 4k = 0$. (2 marks)

Given that $x^2 - 16kx + 4k$ has equal roots,

b. find the solution of the simultaneous equations for this value of k. (6 marks)

Gold

The simultaneous equations
$y + \frac{1}{2} = 5x$
$x^2 - 4ky + 3k = 0$
where k is a non-zero constant, have exactly one pair of solutions.
Find the solution. (8 marks)

Problem solving: Set B

Bronze

$p(x) = x + 6$ and $q(x) = x^2 + 5x + 6$

a. Solve the equation $q(x) = 0$. (2 marks)

b. Sketch the graphs of $y = p(x)$ and $y = q(x)$ on the same set of axes. Label all points where the curves intersect the coordinate axes. (4 marks)

c. Use an algebraic method to find the coordinates of any points of intersection of the graphs $y = p(x)$ and $y = q(x)$. (4 marks)

d. Write the set of values of x for which $p(x) < q(x)$. (2 marks)

Silver

$p(x) = -x - 1.75$ and $q(x) = x^2 + 3x - 18$

a. Sketch the graphs of $y = p(x)$ and $y = q(x)$ on the same set of axes. Label all points where the curves intersect the coordinate axes. (6 marks)

b. Use an algebraic method to find the coordinates of any points of intersection of the graphs $y = p(x)$ and $y = q(x)$. (4 marks)

c. Write the set of values of x for which $p(x) > q(x)$. (2 marks)

d. Shade the region that satisfies the inequalities:
$p(x) < y < q(x), \quad x > 3$ (2 marks)

Gold

$p(x) = 3 + 2x - x^2$, $q(x) = 3x^2 + 2x - 1$

a. Sketch the graphs of $y = p(x)$ and $y = q(x)$ on the same set of axes. Label all points where the graphs intersect each other and the coordinate axes. (10 marks)

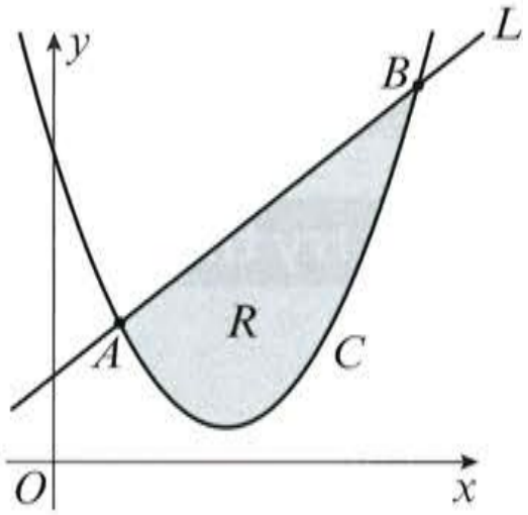
b. Write down, using set notation, the set of values of x for which $p(x) < q(x)$. (2 marks)

c. Shade the region on your sketch which satisfies the inequalities
$y \geqslant p(x)$, $y \geqslant q(x)$ and $x > -1.5$ (3 marks)

Selected exam question bank questions

Question 10

The diagram shows part of curve C with equation $y = x^2 - 8x + 18$ and part of the line L with equation $y = 2x + 5$.



a. Using an appropriate algebraic method, find the coordinates of A and B. (4 marks)
b. Define, using inequalities, the shaded region R. (2 marks)

Question 13

$f(x) = x^2 + (k-4)x + (\frac{1}{2}k + 3)$

a. Find the discriminant of $f(x)$ in terms of k. (3 marks)
b. Given that the equation $f(x) = 0$ has two equal roots, find the possible values of k. (2 marks)
c. Show that when $k = 8$, $f(x) > 0$ for all values of x. (3 marks)

Question 23

The equation $kx^2 + 5kx + 5 = 0$, where k is a constant, has no real roots. Prove that $0 < k < \frac{4}{5}$. (4 marks)

Question 32

Find the set of values of x for which:

a. $3x - 7 > 3 - 7x$ (2 marks)
b. $x^2 - 4x \leq 21$ (4 marks)
c. **both** $3x - 7 > 3 - 7x$ **and** $x^2 - 4x \leq 21$. (1 mark)

Question 35

Solve the simultaneous equations

$$x + y = 2 \qquad 3y^2 - x^2 = 12$$

(6 marks)

Question 55

The straight line with equation $y = 2x - 4$ does not cross or touch the curve with equation $y = x^2 + px - 2p$, where p is a constant.

a. Show that $p^2 + 4p - 12 < 0$. (3 marks)

b. Hence find the set of possible values of p. (3 marks)

Question 62

The line with equation $mx - y - 3 = 0$ touches the circle with equation $x^2 - 4x + y^2 - 6y = 7$. Find the two possible values of m. (7 marks)

Question 65

The line $y = k - x$ is a tangent to the curve $y = x^2 + 5x - 2$. Find the value of k. (3 marks)

Question 71

a. Show that $x^2 - 8x + 17 > 0$ for all real values of x. (3 marks)

b. "If I add 2 to a number and square the sum, the result is greater than the square of the original number."
State, giving a reason, whether the above statement is always true, sometimes true or never true. (2 marks)

Answers

3.1 Linear simultaneous equations

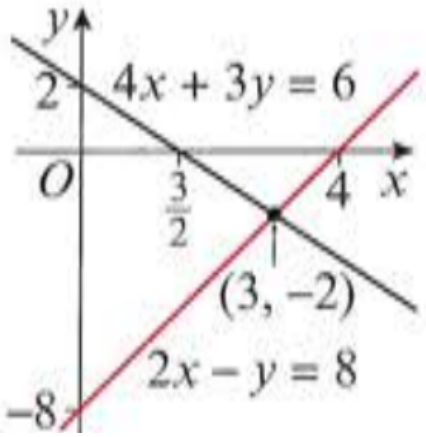
1. **a** $x = 2$, $y = -3$ **b** $x = 1$, $y = 2$ **c** $x = -1$, $y = -2$
2. **a** $x = 1$, $y = 1$ **b** $x = 2.25$, $y = 6.5$ **c** $x = 0.5$, $y = 1$
3. **a** $x = 3$, $y = 5$ **b** $x = 3$, $y = -5$ **c** $x = 5$, $y = -2$
4. $x = 0.5$, $y = 1.5$
5. $x = \frac{3}{7}$, $y = \frac{44}{7}$
6. **a.** $(6ky + 9x = 12) - (6ky - 6x = 27)$ gives $15x = -15$, so $x = -1$
   b. $k = 0.5$
7. Nisha. You can rearrange each equation to get the other, so any pair of values that satisfies one equation will satisfy both.

3.2 Quadratic simultaneous equations

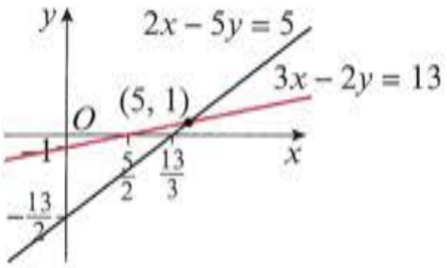
1. **a.** $x = -1$, $y = -64$ or $x = 16$, $y = 4$
   b. $x = 1$, $y = 3$ or $x = 3$, $y = 1$
   c. $x = 2$, $y = 3$ or $x = -\frac{2}{3}$, $y = \frac{1}{3}$
2. **a.** $x = 1$, $y = -4$ or $x = 4$, $y = 2$
   b. $x = -4$, $y = -6$ or $x = 3$, $y = 1$
   c. $x = 1$, $y = 2$ or $x = 2$, $y = -1$
3. **a.** $x = 1.63$, $y = -2.12$ or $x = 7.37$, $y = 15.12$
   b. $x = -1.00$, $y = 2.00$ or $x = 1.75$, $y = -0.75$
   c. $x = 0.59$, $y = -4.66$ or $x = 4.51$, $y = 11.02$
4. $x = 2$, $y = 2$ or $x = 8.67$, $y = -4.67$ (3 s.f.)
5. $x = -0.631$, $y = 0.261$ or $x = -0.227$, $y = -0.547$ (3 s.f.)
6. $\dfrac{-2 \pm 3\sqrt{2}}{4}$
7. **a.** $y = 8 + 2x \Rightarrow x^2 + 2k(8 + 2x) + 4k = 0 \Rightarrow x^2 + 4kx + 20k = 0$
   b. $k = 5$
   c. $x = -10$, $y = -12$

3.3 Simultaneous equations on graphs

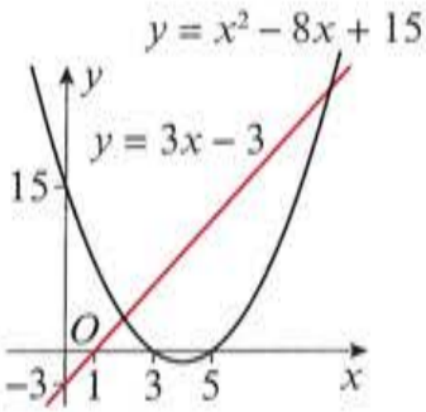
1. **a.**



b.



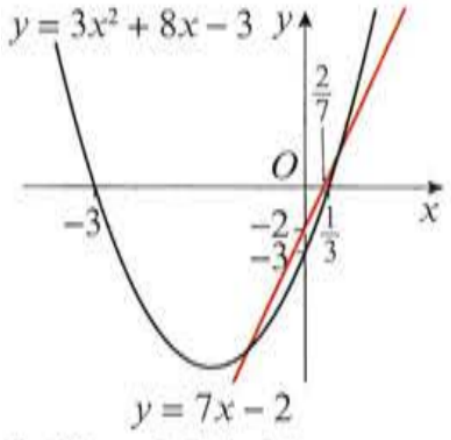
2. **a.**



b. $(2, 3)$ and $(9, 24)$

c. $x = 2:\ 2^2 - 8 \times 2 + 15 = 3 \times 2 - 3 = 3$
$x = 9:\ 9^2 - 8 \times 9 + 15 = 3 \times 9 - 3 = 24$

3. **a.**

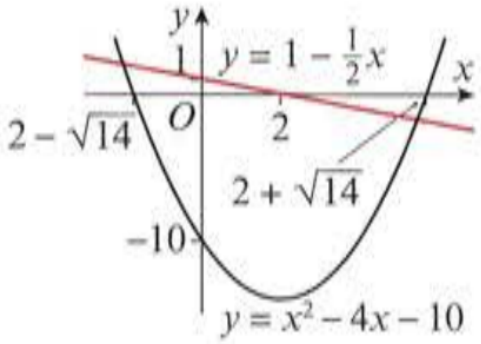


b. $(-0.77, -7.37)$ and $(0.43, 1.04)$

4. $A(1, 13)$ and $B(4, 16)$

5. $A(-2, 5)$ and $B(2, -3)$

6. **a.**



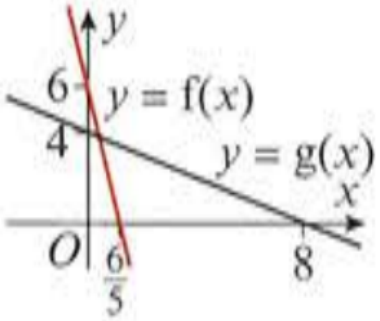
b. $(-2, 2)$ and $(5.5, -1.75)$

7. $x = 5.18$ (3 s.f.)

8.

a. $(\frac{4}{9}, \frac{34}{9})$

b.



3.4 Linear inequalities

1. **a** $x \geqslant 8$ **b** $x > -11$ **c** $x \geqslant \frac{3}{10}$

2. **a** $2 \leqslant x \leqslant 4$ **b** $x < 27$ **c** $x \geqslant \frac{1}{3}$

3.
- **a.** $\{x : -3 < x < \frac{2}{3}\}$
- **b.** $\{x : 5 \leqslant x \leqslant 12\}$
- **c.** $\{x : x < \frac{1}{2}\} \cup \{x : x > 38\}$

4. **a** $x > 3$ **b** $x < 4$

5. **a** $\{x : x > -1\}$ **b** $\{x : x \leqslant \frac{4}{5}\}$

6. **a** $\{x : x \geqslant -1\}$ **b** $\{x : x < 3\}$

7. **a** $2(x + 30) + 2x < 400$ **b** $x < 85$ m

8.
- **a.** $\{x : -1 \leqslant x < 2\}$
- **b.** $\{x : x < 2.25\} \cup \{x : x > 101\}$

3.5 Quadratic inequalities

1. **a** $-5 < x < 5$ **b** $-\frac{3}{2} < x < 1$
c $x < -1$, $x > \frac{1}{3}$ **d** $x \leqslant 0$, $x \geqslant \frac{5}{2}$

2.
- **a.** $\{x : 2 < x < 7\}$
- **b.** $\{x : -4 < x < -1\}$
- **c.** $\{x : -5 < x < -3\} \cup \{x : -2 < x < 1\}$

3. **a** $x < -\frac{1}{6}$, $x > \frac{1}{6}$ **b** $0 < x < \frac{5}{3}$
c $x \leqslant -\frac{1}{4}$, $x > 0$

4. $x^2 - 6x + 11 = (x - 3)^2 + 2 \Rightarrow x^2 - 6x + 11 \geqslant 2 > 0$

5. **a** $x > 3$ **b** $-4 \leqslant x \leqslant 10$ **c** $3 < x \leqslant 10$

6.
- **a.** $(k - 2)^2 - 4 \times 1 \times (4 - 2k) > 0$, therefore $k^2 + 4k - 12 > 0$
- **b.** $k < -6$, $k > 2$

7.
- **a.** $22x + 4 > 103 \Rightarrow x > 4.5$
- **b.** $21x^2 + 9x < 702 \Rightarrow -6 < x < \frac{39}{7}$
- **c.** $4.5 < x < \frac{39}{7}$

8. $0 < k < \frac{8}{9}$

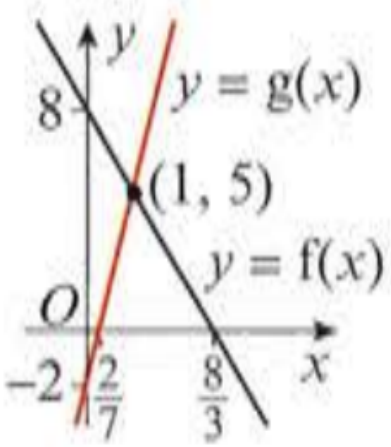
9. $\dfrac{10 - 2\sqrt{7}}{9} < p < \dfrac{10 + 2\sqrt{7}}{9}$

10. $x < 1$ or $x > 2.5$

3.6 Inequalities on graphs

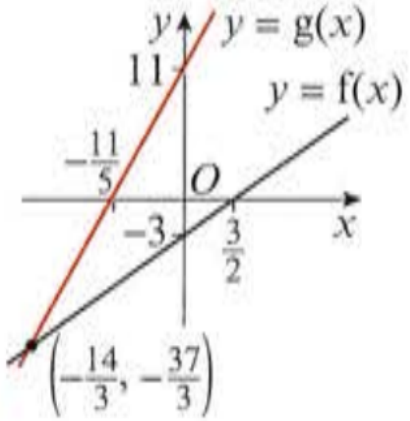
1.

a. i



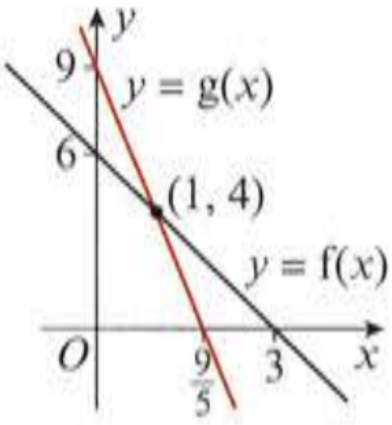
ii $x > 1$

b. i



ii $x > -\frac{14}{3}$

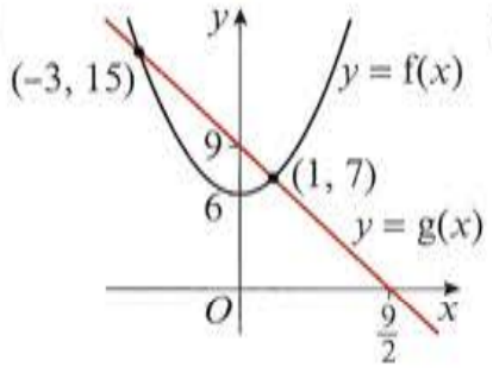
c. i



ii $x < 1$

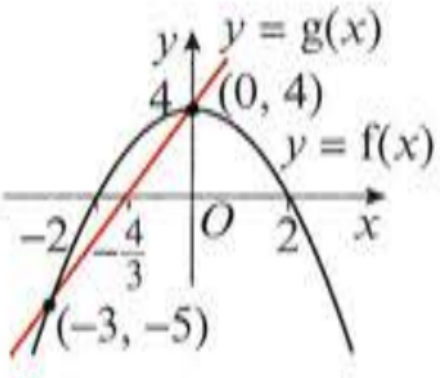
2.

a. i



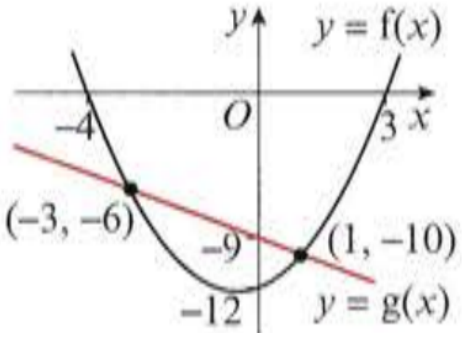
ii $-3 \leqslant x \leqslant 1$

b. i



ii $x \leqslant -3$ and $x \geqslant 0$

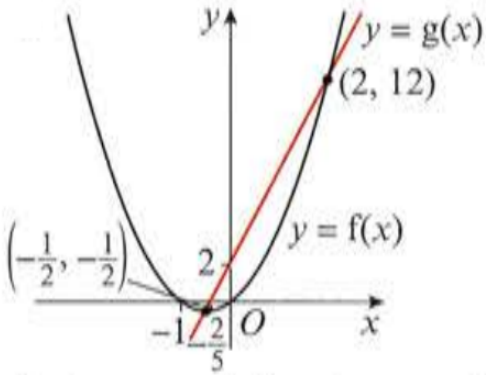
c. i



ii $-3 \leqslant x \leqslant 1$

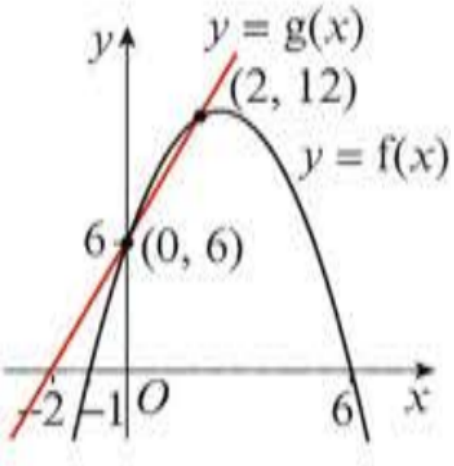
3.

a. i



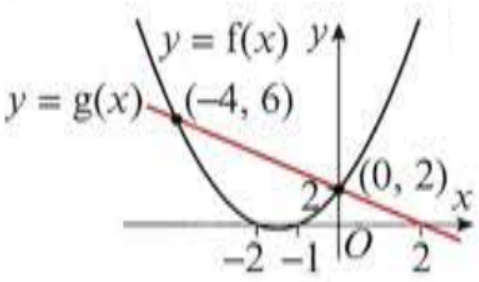
ii $\{x : x < -0.5\} \cup \{x : x > 2\}$

b. i



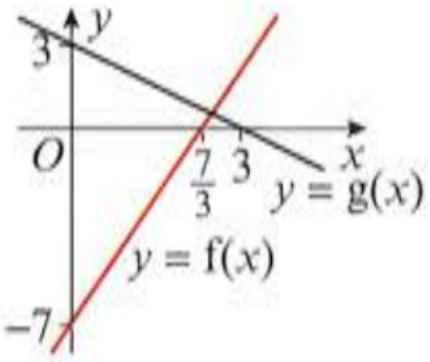
ii $\{x : 0 < x < 2\}$

c. i



ii $\{x : x < -4\} \cup \{x : x > 0\}$

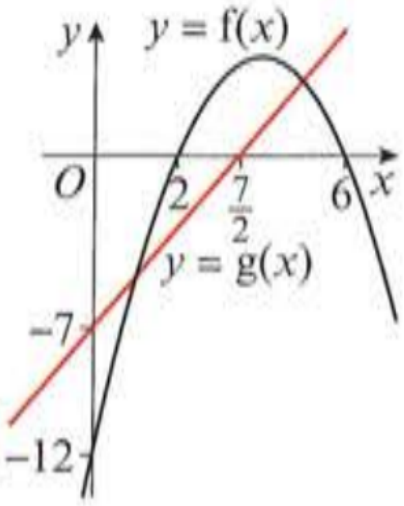
4. **a.**



b. $(\frac{5}{2}, \frac{1}{2})$

c. $x > \frac{5}{2}$

5. **a.**

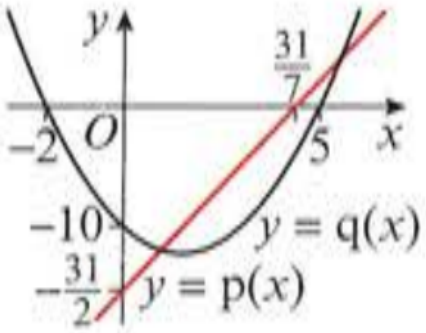


b. $(1, -5)$ and $(5, 3)$

c. $1 < x < 5$

6. **a.** $x = -2$ and 5

b.

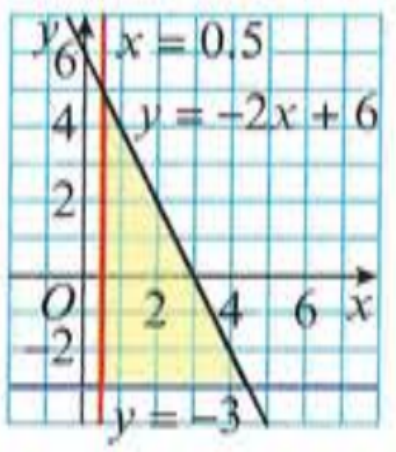


c. $(1, -12)$ and $(\frac{11}{2}, \frac{15}{4})$

d. $\{x : x < 1\} \cup \{x : x > \frac{11}{2}\}$

3.7 Regions

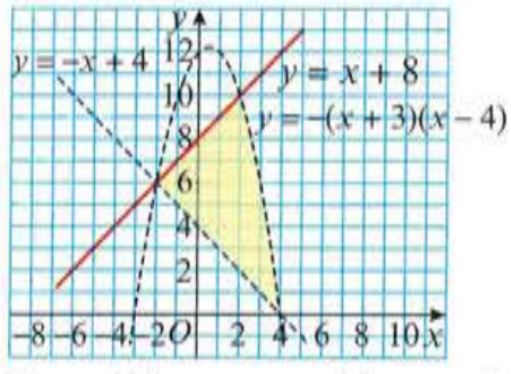
1. **a.**



b. $(0.5, 5)$, $(4.5, -3)$ and $(0.5, -3)$

c. 16

2. **a.**



b. $(4, 0)$, $(-2, 6)$, $(2, 10)$

c. They all lie on one of the two lines with strict inequalities.

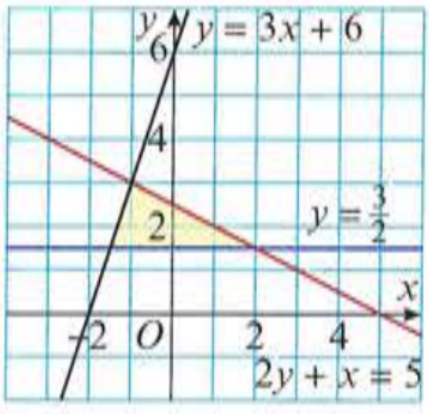
3.

a. $(0, 2)$, $(-1, 0)$, $(-1.5, 0.5)$ and $(-2, 0)$

b. $y \geqslant x^2 + 3x + 2$, $y \geqslant 2x^2 + 4x + 2$ and $y \geqslant x + 2$

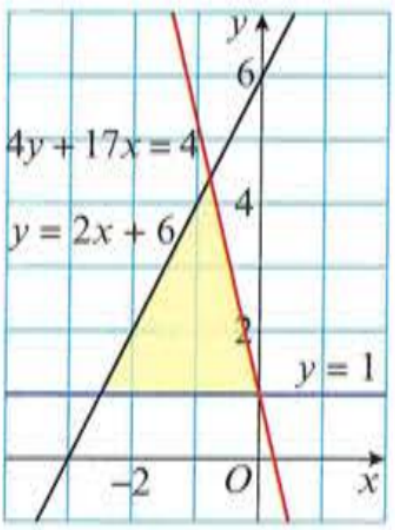
c. $(-1.5, 0.5)$ and $(0, 2)$

4. **a.**



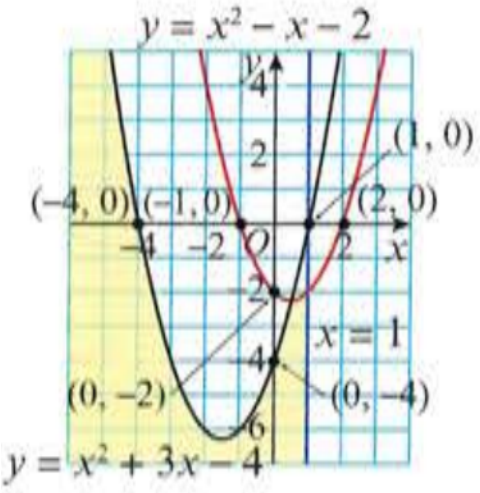
b. $\frac{21}{8}$

5. **a.**



b. $\frac{17}{4}$

6. **a, c**



b $(0.5, -2.25)$, $(1, 0)$ and $(1, -2)$

7. **a** 6.25 **b** $y \geqslant 2\sqrt{x} - 3$, $y \leqslant 10 - 1.28x$, $x \geqslant 2.25$

Problem solving: Set A

Bronze

a $x^2 - (2x + 3) + 2k = 0 \Rightarrow x^2 - 2x + (2k - 3) = 0$

b $k = 2$ **c** $x = 1$, $y = 5$

Silver

a $x^2 - 4k(4x - \frac{1}{4}) + 3k = 0 \Rightarrow x^2 - 16kx + 4k = 0$

b $x = \frac{1}{2}$, $y = \frac{7}{4}$

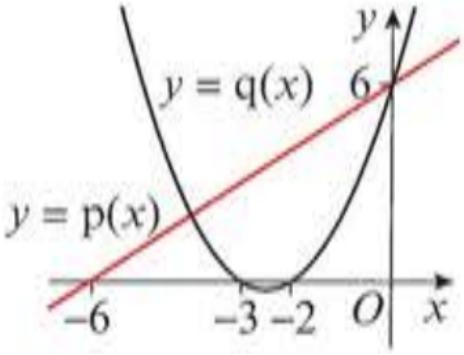
Gold $x = \frac{1}{2}$, $y = 2$

Problem solving: Set B

Bronze

a $x = -2$ and $x = -3$

b

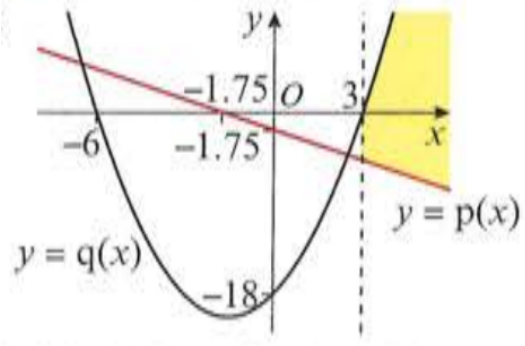


c $(-4, 2)$ and $(0, 6)$

d $x < -4$ or $x > 0$

Silver

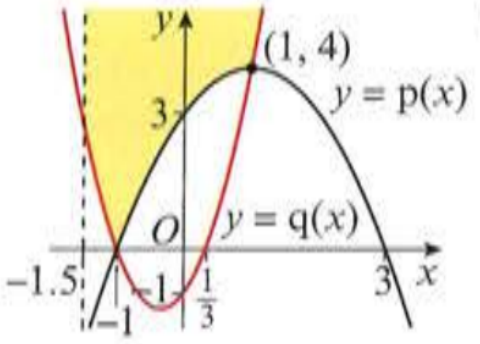
a, d



b $(-6.5, 4.75)$ and $(2.5, -4.25)$ **c** $-6.5 < x < 2.5$

Gold

a, c



b $\{x : x < -1\} \cup \{x : x > 1\}$

Selected exam question bank answers

Question 10

a. $A(5-2\sqrt{3}, 15-4\sqrt{3})$, $B(5+2\sqrt{3}, 15+4\sqrt{3})$

b. $x^2-8x+18 \leq y \leq 2x+5$

Question 13

a. $k^2-10k+4$

b. $k = 5 \pm \sqrt{21}$

c. Substituting $k=8$ into $f(x)$ gives $f(x) = x^2+4x+7 = (x+2)^2+3$. So $f(x) > 0$ for all values of x.

Question 23 $25k^2-20k < 0 \Rightarrow 5k(5k-4) < 0 \Rightarrow 0 < k < \frac{4}{5}$

Question 32

a. $x > 1$

b. $-3 \leq x \leq 7$

c. $1 < x \leq 7$

Question 35 $x=0$, $y=2$ or $x=6$, $y=-4$

Question 55

a. $2x-4 = x^2+px-2p \Rightarrow x^2+(p-2)x+4-2p=0$
So $(p-2)^2-4(4-2p) < 0 \Rightarrow p^2+4p-12 < 0$

b. $-6 < p < 2$

Question 62 $\frac{1}{2}$ and -2

Question 65 $k=-11$

Question 71

a. $x^2-8x+17 = (x-4)^2+1$. Minimum value is $y=1$.

b. Sometimes true; true if $x > -1$, else false.